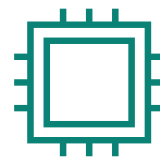




Boosting OEM Profitability: AURIX™ TC4x PPU Transforms Your E-Drivetrain

Dr. Renke Han – Principal System Architect

Embedded World 2026



Market leader
MCU



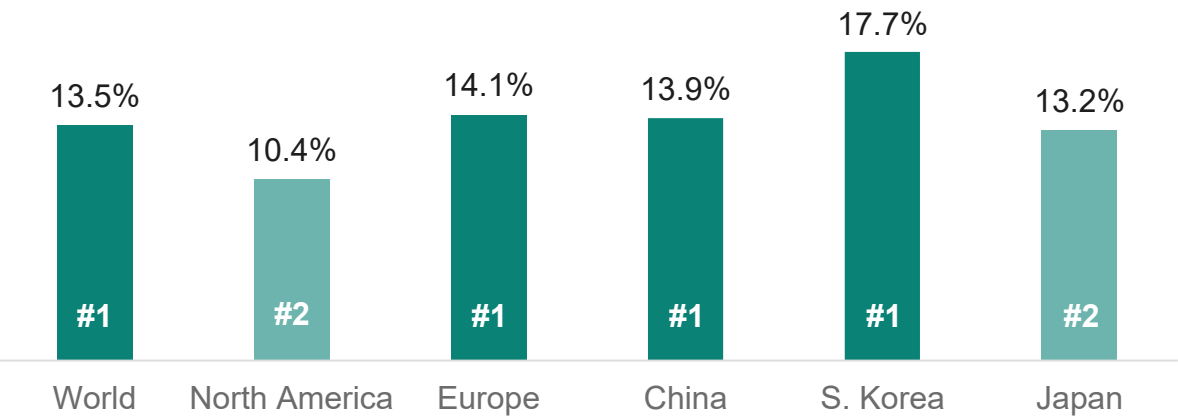
Infiniteon's top market position is built on system competence based on an industry-leading product portfolio



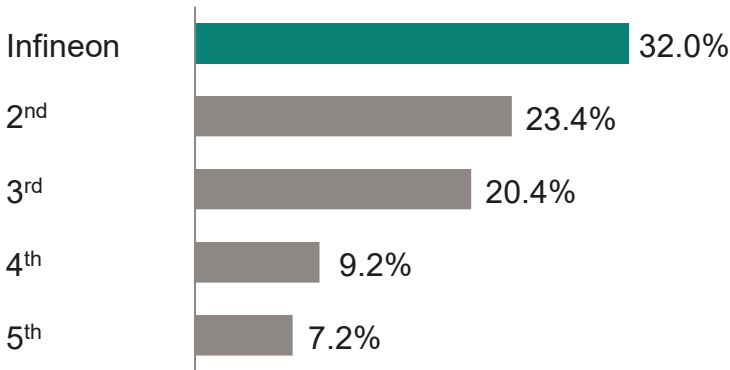
Automotive semiconductors (2024 total market: \$68,382m; -1.2% y-y)



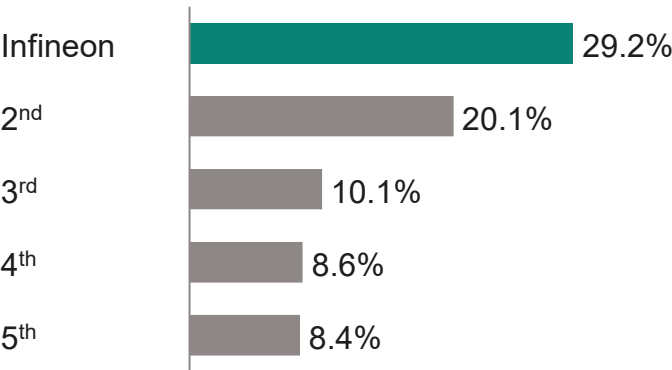
Infineon's 2024 market share and position by region



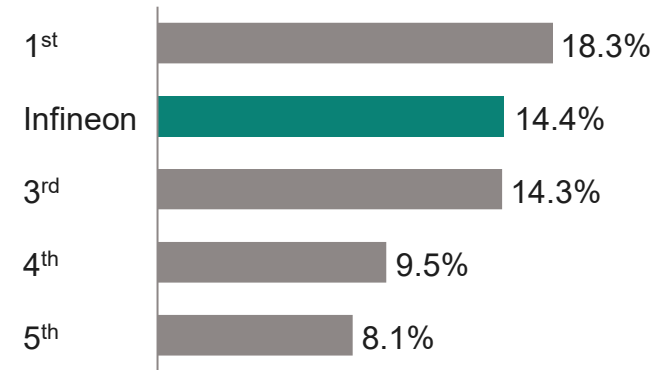
MCUs



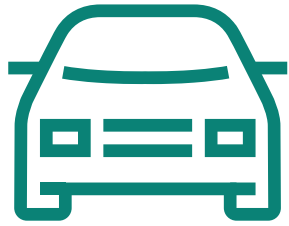
Power semiconductors



Sensors



TechInsights: Automotive Semiconductor Vendor Market Shares. March 2025. Sensors: S&P Global Mobility: Automotive Semiconductor Market Share Database. May 2025.



**Two perspectives boosting
OEM profitability**



System Cost down!
System Cost down!!
System Cost down!!!



Willingness-to-Pay
features

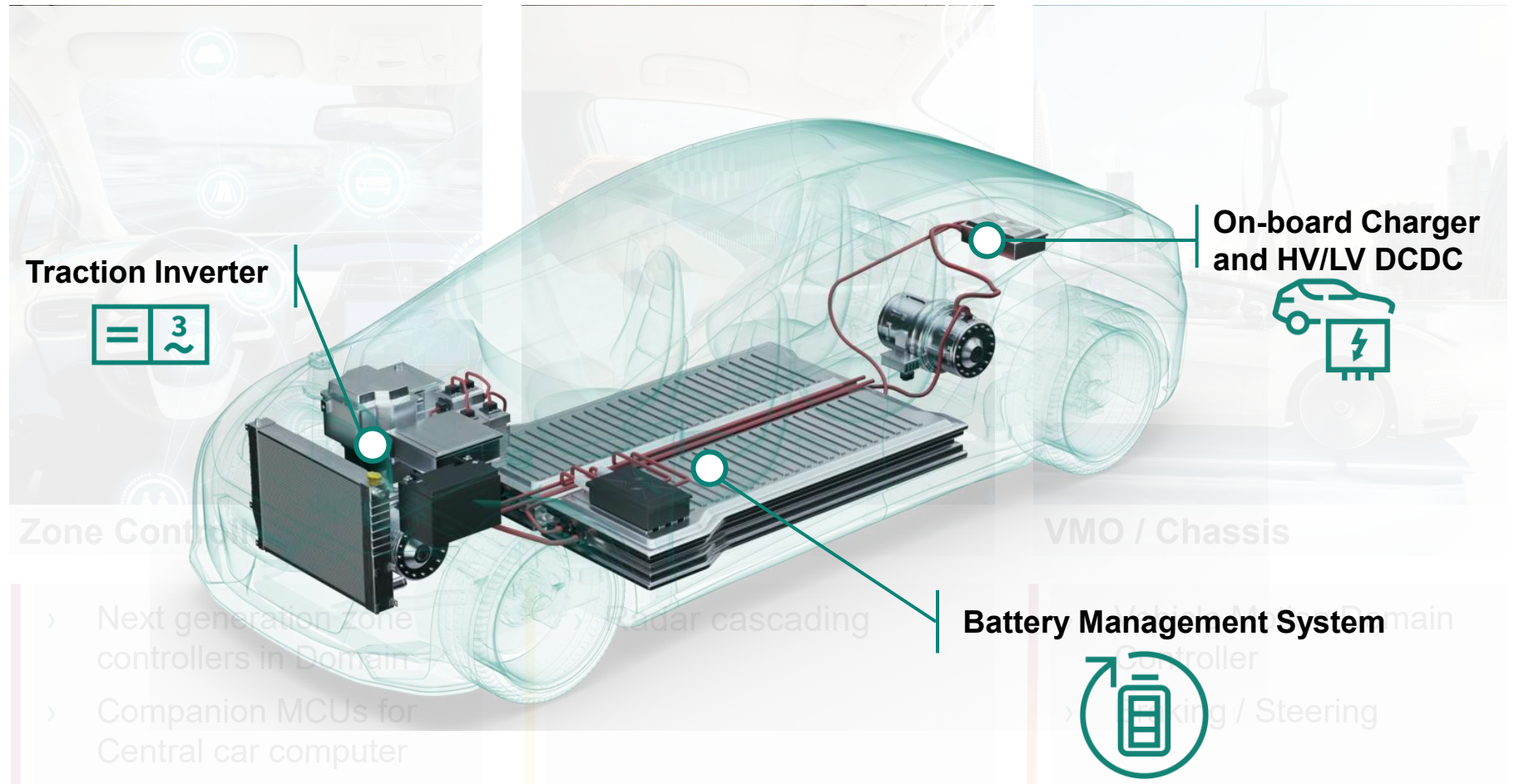


AURIX™ TC4x comprehensive portfolio tailored serving major market trends in our focus applications



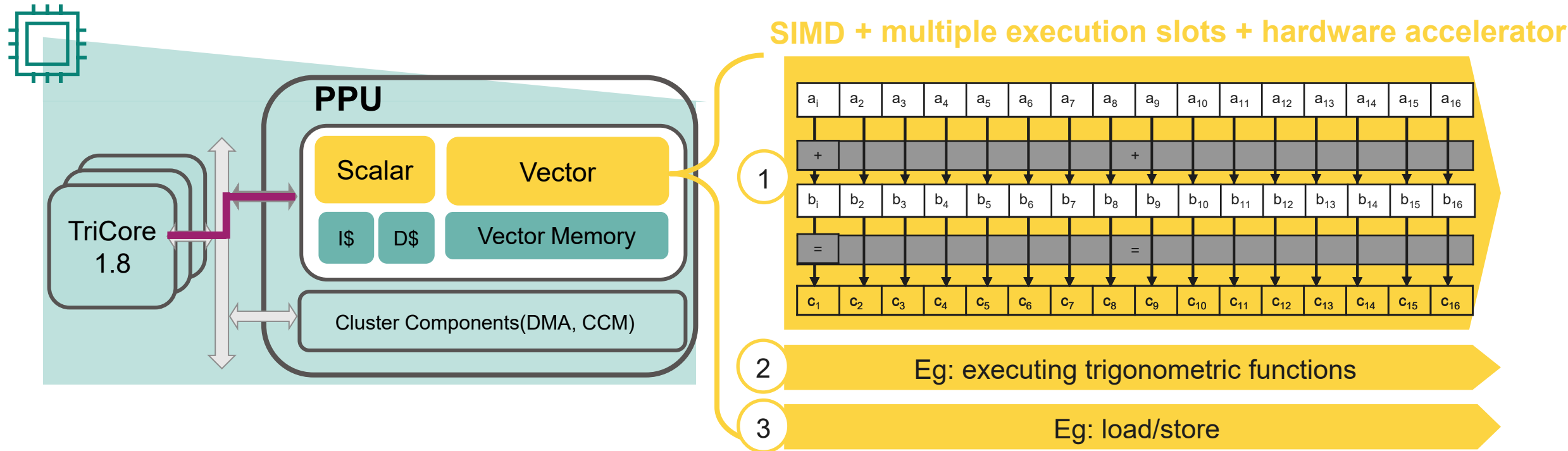
Electrification

- › Traction Inverter
- › Power Conversion
- › Battery Management System



AURIX™ TC4x Parallel Processing Unit (PPU) goes beyond an AI accelerator and does more than integer-based SIMD*

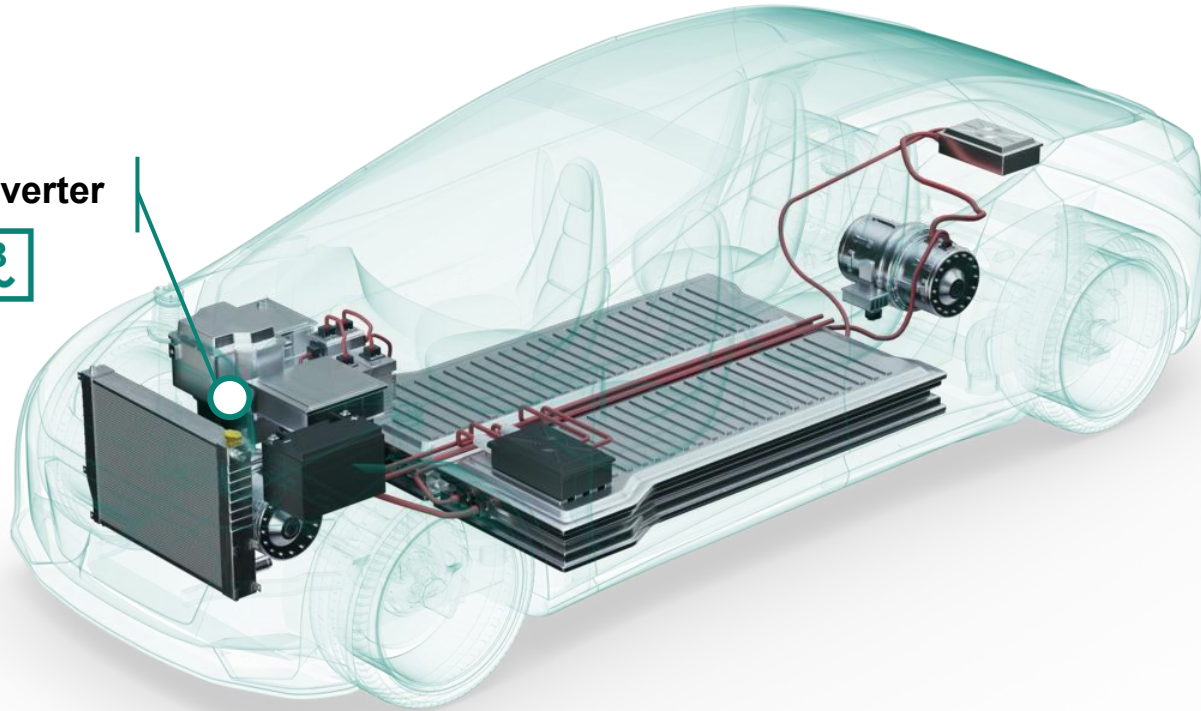
*SIMD: single instruction multiple data



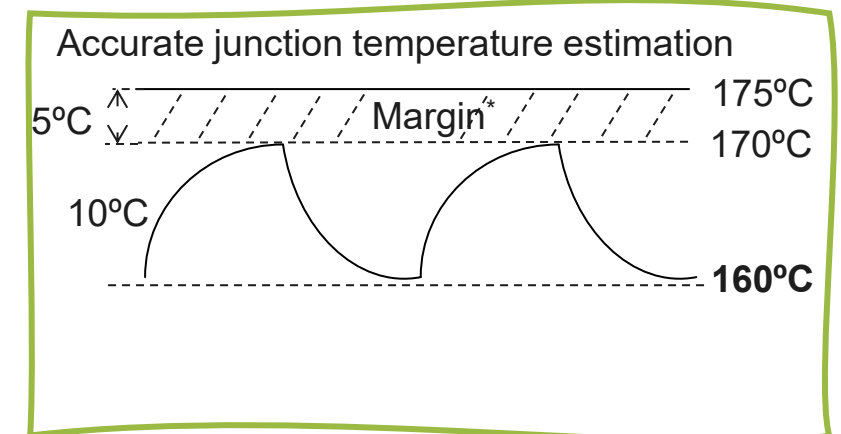
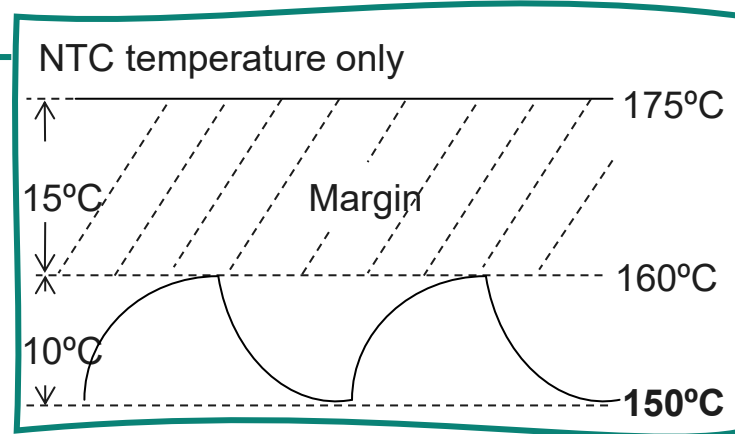
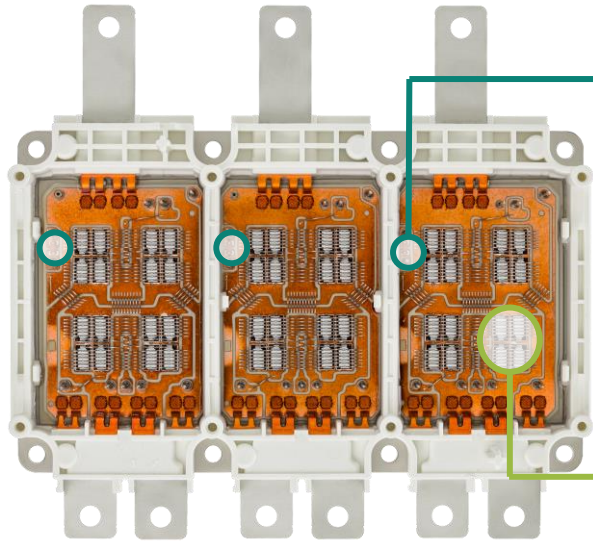
PPU is a dedicated compute core with **SIMD and multiple execution slots capabilities**, enabling N times faster than competitors, e.g., floating matrix multiplication-based algorithms and parallel High-speed-control loops

Virtual sensing helps traction inverter achieving system cost down

Traction Inverter



AURIX™ TC4x PPU allows accurate junction temperature estimation in real-time enabling double-digit system cost savings



Assumptions: * Temperature accuracy: +/-2 Kelvin

Two ways of junction temperature calculation

Traditional:
Thermal impedance based
(one paper at PCIM@2026)

vs

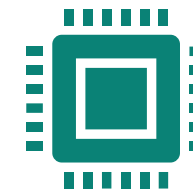
New:
AI based

Both methods strongly involve floating **matrix calculation**,
hence PPU can overtake the calculation in real-time (xx
us) during operation

Remove 10°C margin

7% die size saving

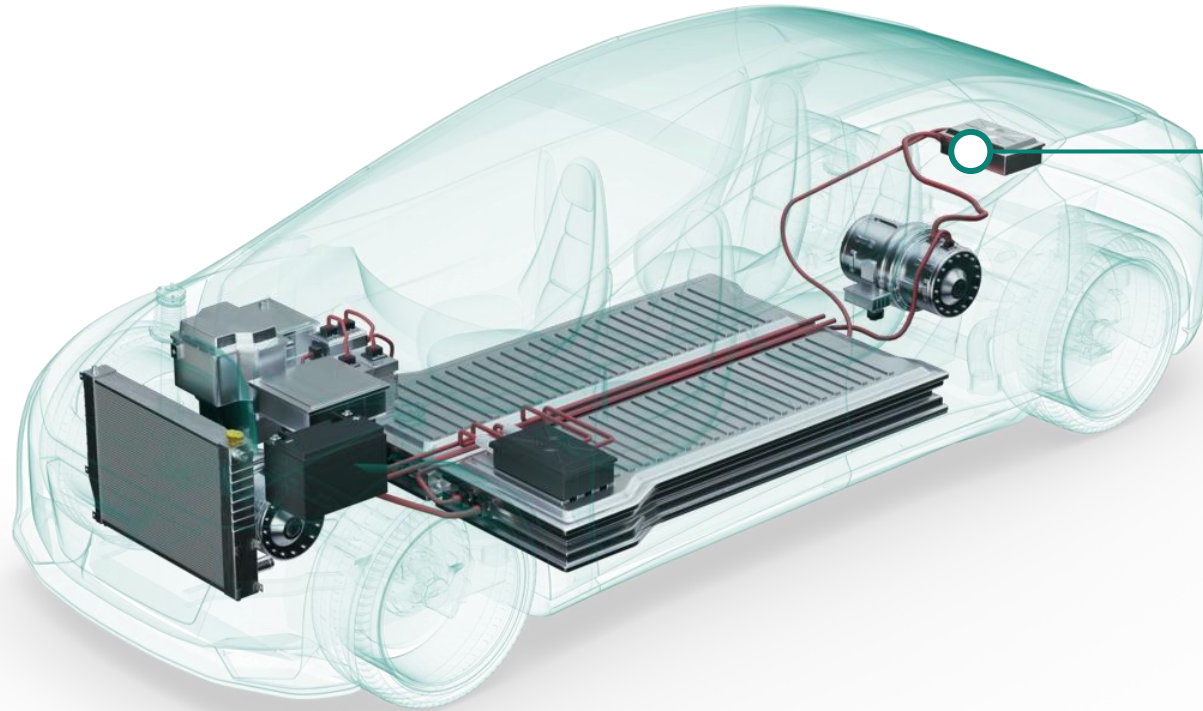
5% power module saving**



>10 EUR system cost
savings on power modules

Assumptions: ** 70% cost of power module is from the chip

Integration solution for OBC+DCDC clearly drives system cost down



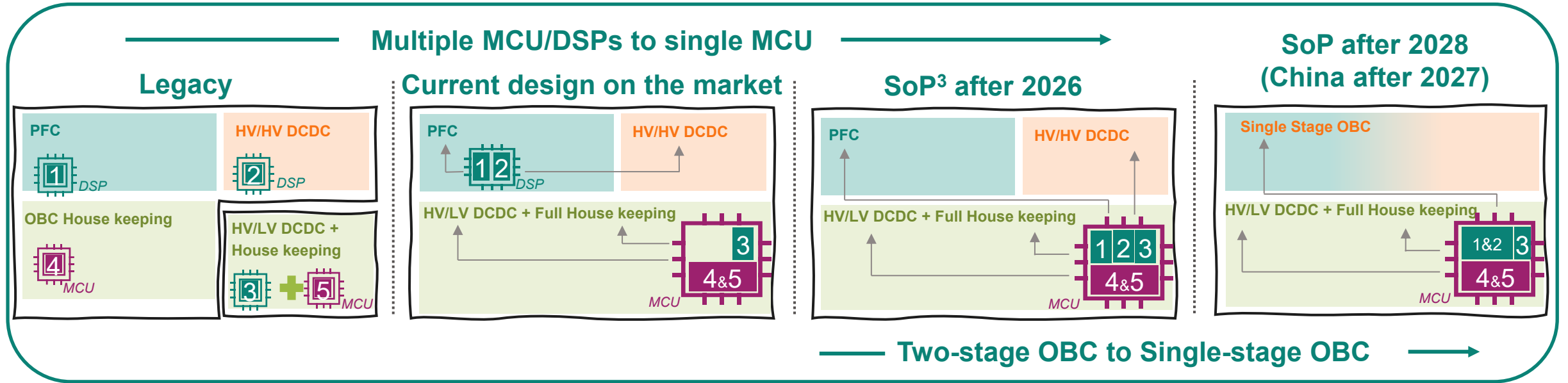
On-board Charger
and HV/LV DCDC



OBC/DCDC market evolution paths are from multiple MCUs/DSPs to single MCU, then from Two-stage to Single-stage OBC

Legend  ECU¹  DSP²  MCU  House keeping  Control loop

¹ ECU: Electronics control unit ³ SoP: Start of Production
² DSP: digital signal processor



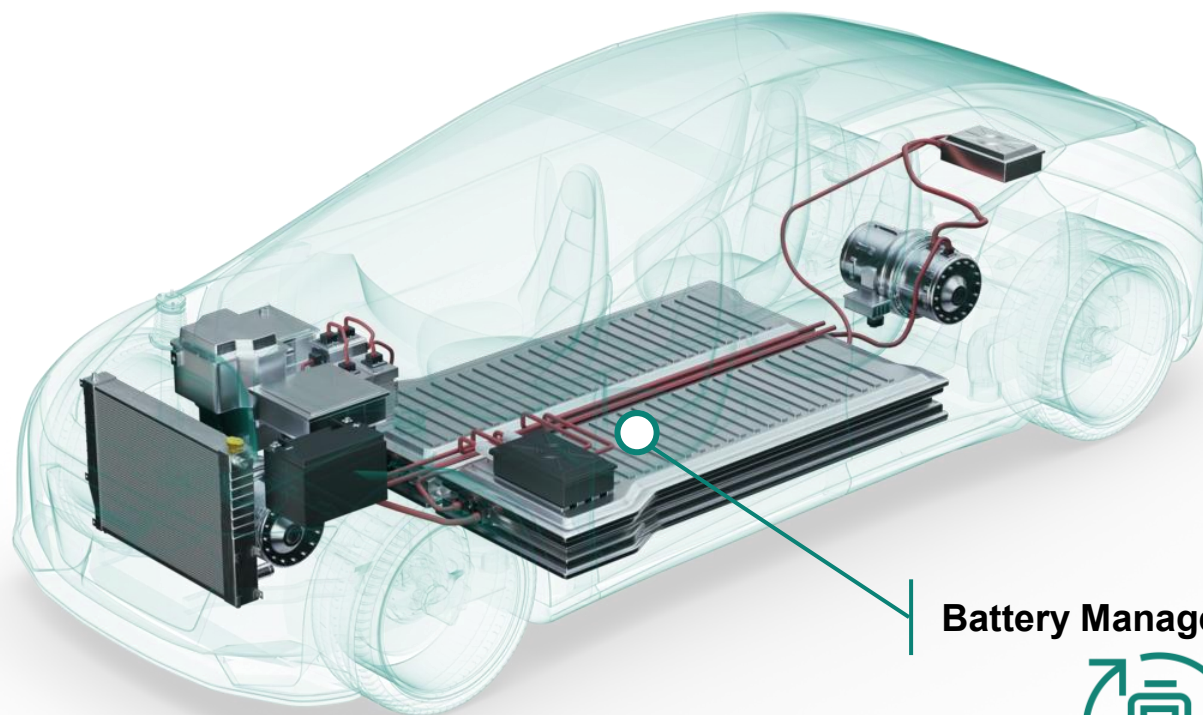
AURIX™ TC4x PPU enable **parallel fast control loops (e.g., 2μs)** saving ≥ 1 application core, and its optimized real-time cluster allows for coordinated control among stages improving overall system efficiency

— **Single** firmware maintenance, **less** licenses, **one** part number etc.....

— **Remove** 1-2 PCBs and several automotive qualified connectors

— Reinforced isolation chips getting cheaper while **bringing 20 years application lifetime**⁴

The next level of estimation accuracy enables faster charging, alleviates range anxiety, and increases Willingness-to-Pay values



Battery Management System



Increasing functional requirements and the accuracy of BMS functions are adding Willingness-to-Pay values for OEMs

BMS in the past ...

... primarily focused on safety

- **Static** algorithms
- State estimation as “comfort functions” with **huge safety margins**
- **Simple**, easy-to-develop **models**, parametrized based on empirical data (e.g., aging tests)



- Range anxiety
- Uncertain resale values
- Safety concerns

Increasing importance of the BMS

Nowadays, the BMS ...

... require more precise model to monitor battery under all operating conditions through its service life

- Future requirements for **mandatory battery information**¹:
 - Anode potential estimation², Lithium plating detection
 - Capacity fade (SoH) – below 4%
 - ...
- Quality of state indicators are of **high relevance for customer** (esp. throughout service life)
- Optimal utilization to maximize performance of valuable cells is **key differentiator for OEMs**



- Remove range anxiety by enabling fast charging
- More accurate SoX guarantees battery resale value
- Effective algorithms (eg: Lithium plating detection) enable predictive maintenance

AURIX™ TC4x PPU...

- ... enables running **precise electro-chemical models** to replace equivalent circuit model in real-time
- ... enables execution **precise algorithms in parallel** (eg: lithium plating detection, anode potential estimation²) monitoring HV battery cells (100+ cells)

¹ UN GTR No. 22, Battery passport consortium: https://thebatterypass.eu/assets/images/content-guidance/pdf/2023_Battery_Passport_Content_Guidance.pdf
² <https://www.sciencedirect.com/science/article/pii/S037877532400572X>

AURIX™ TC4x PPU helps OEMs improve profitability by optimizing costs and enabling willingness-to-pay features

- Motor Position Sensing
- Health Observing
- Model Predictive Control
- **Tj temperature estimation**

- Edge AI
- **Integration**

Traction Inverter

On-board charger and HV/LV DCDC

Battery Management system

- **Advanced model for SoC/SoH**
- Electro impedance spectroscopy (EIS)
- Lifetime estimation

